SIEMENS

Installation Instructions
Model FDBZ492-PR

Global Air Duct Monitoring Housing

INTRODUCTION

Model FDBZ492-PR from Siemens Industry, Inc., is a RoHS compliant air duct monitoring housing containing sampling tubes. When used with a Siemens conventional smoke detector, smoke and combustion products are detected for shutdown of the duct system and/or operation of supplementary equipment as provided by the system control panel. The duct unit supports two sets of Form C alarm contacts, one Form A alarm contact and one Form C trouble contact. The trouble contact supervises the presence of the input power, removal of the detector cover and the removal of the smoke detector head.

TECHNICAL DATA

AIR DUCT CONDITIONS

Temperature Range:
With PE-11, 8854, PE-11C: 0°C (32°F) - 38°C (100°F)
With OP121: 0°C (32°F) - 49°C (120°F)
Altitude Range:
No altitude limitations
Relative Humidity:
Up to 95% RH (non-condensing/non-freezing)
Air Duct Velocity Range:
100-4000 ft/min
Sampling and Exhaust Tube Pressure Differential Range:
Greater than 0.01 and less than 1.2 inches of water column



These air duct detectors are designed for detection and control of products of combustion in a duct system. They are not to be used as a substitution for open area protection.

DO NOT USE air duct detectors with Alarm Verification.

ASSEMBLY

This detector has a cover tamper (removal) switch. Care should be taken when installing the cover. Squarely place the cover on the unit to aviod possible damage to the switch.



DO NOT SLIDE COVER INTO POSITION.

P/N A6V10330327_a_en

Siemens Industry, Inc.
Building Technologies Division

APPLICATION

The FDBZ492-PR duct smoke detectors provide early detection of smoke and products of combustion present in air moving through an HVAC duct supply, return, or both. These devices are designed to prevent the recirculation of smoke in areas by the air handling system's fans and blowers and may be used with fire alarm systems that shut down complete air handling system in the event of smoke detection.

NOTE

For the correct installation of a duct smoke unit please refer to NFPA 72 (National Fire Alarm Code), NFPA 90A (Standard for Installation of Air Conditioning and Ventilation Systems), and NFPA 92A (Recommended Practice for Smoke Control Systems).

This unit is not intended for open area protection nor should it be used for early warning detection or to replace a regular fire detection system.

OPERATION

When the Model FDBZ492-PR is operating, a sample of air is drawn from the duct and passed through the sampling chamber by means of the input sampling tube. The air sample passes through the smoke detector mounted in the duct housing and is exhausted back into the duct through the outlet tube.

- The FDBZ492-PR supports two sets of Form C alarm contacts, one Form A alarm contact and one Form C trouble contact. The trouble contact supervises the presence of the input power, removal of the detector cover and the removal of the smoke detector head.
- The FDBZ492-PR contains a green power on LED, red alarm LED and yellow trouble LED which are visible through the housing cover.
- The trouble LED is activated when either the housing cover or smoke detector is removed from the unit.
- When the trouble LED is activated, the power LED turns off.
- A reset/test switch is located on the housing cover and is used to reset the unit after an alarm and the smoke condition is cleared. When not in alarm, the reset/test switch can be used to test that the unit will operate properly when an alarm condition exists.

The trouble contacts will not operate in the event of a smoke alarm. The FDBZ492-PR duct detector will operate from various input voltage sources; namely 24VAC, 24VDC, 115VAC and 230VAC.

Detector LED

The FDBZ492-PR contains an LED indicator (located on the smoke detector) capable of flashing either one of three distinct colors: green, yellow, or red. During each flash interval, the microprocessor based detector checks the following:

- for smoke in its sensing chamber
- that its critical smoke sensing electronics are operating.

Based on the results of these checks, the LED indicator flashes as follows:

Detector	Flash Color	Condition	Flash Interval (Seconds)
PE-11 8854 PE-11C	Green	Normal supervisory operation.	7
OP121	Green	Normal supervisory operation.	10
PE-11 8854 PE-11C	Yellow	Detector requires service (cleaning or repair) or is operating beyond its environmental specifications.	7-30
OP121	Yellow	Detector is in trouble and needs replacement.	5
PE-11 8854 PE-11C OP121	Red	Alarm	2½
PE-11 8854 PE-11C	No Flashes	Detector is not powered or requires repair.	__
OP121	No Flashes	Detector is not powered or replacement is needed.	__

MOUNTING THE AIR DUCT HOUSING

Location on Duct System

This guideline contains general information on duct smoke detector installation, but does not preclude the NFPA documents listed. Siemens Industry, Inc. assumes no responsibility for improperly installed duct detectors. To determine the correct installation position for an FDBZ492-PR duct smoke detector, the following factors must be considered.

1) A uniform non-turbulent (laminar) airflow between 100 ft/min. to 5,000 ft/min. must be present in the HVAC duct. To determine duct velocities, examine the engineering specifications that define the expected velocities or use an Alnor model 6000AP velocity meter (or equivalent).
2) To minimize the impact of air turbulence and stratification on performance, a duct smoke detector should be located as far as possible downstream from any obstruction (i.e. deflector plates, elbows, dampers, etc.). In all situations,

confirmation of velocity and pressure differential within specifications is required.

The pressure differential between the input sampling (high pressure) tube and outlet (low pressure) tube for the FDBZ492-PR smoke duct detector should be greater than 0.01 inches of water and less than 1.2 inches of water.

3) Identify a code compliant location (supply or return side, or both) for the installation of the duct unit that will permit easy access for viewing and serviceability.

4) When installing on the return side, install duct units prior to the air being exhausted from the building or diluted with outside "fresh" air.

5) When installing duct smoke units downstream of filters, fires occurring in the filters will be detected, but if the filters become blocked, insufficient air flow through the duct unit will prevent the correct operation of the duct detector. Duct units installed in the supply air side may monitor upstream equipment and/or filters.

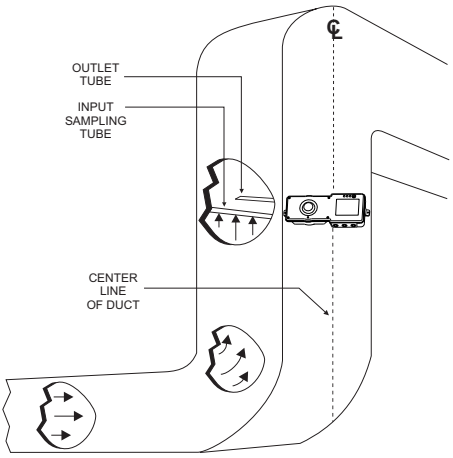


Figure 1
Typical Mounting of Duct Housing

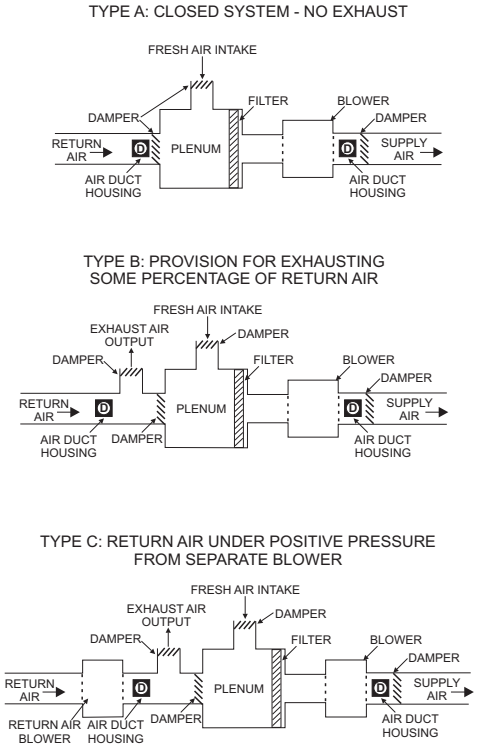


Figure 2
Recommended Locations in Duct Systems

6) Where possible, install duct detectors upstream of air humidifiers and downstream of dehumidifiers.

7) To prevent false alarms, the duct detector should not be mounted in areas of extreme high or low temperatures, in areas where high humidity exists, or in areas where the duct may contain gases or excessive dust.

Duct Preparation

The FDBZ492-PR Housings come with an installation kit that contains the following items:

- Short return (outlet) tube
- Input sampling tube stopper
- Two #12 x 3/4" sheet metal screws
- Mounting template (packaged separately)

Remove mounting template. Remove paper backing from the mounting template and affix it to the duct at the desired location. Using the template as a guide, drill (2) mounting holes, 3/32" (2.5mm) for the #12 X 3/4" sheet metal screws packaged in the installation kit. Drill or punch (2) 1¼" (32mm) holes for input sampling and outlet tubes, using the template as a guide. Clean all holes.

Sampling Tube Installation

The FDBZ492-PR duct smoke detectors use a specially notched sampling tube, which may be ordered separately in one of four standard lengths.

ST-10	For duct widths of 6" to 1.0'	
ST-25	For duct widths of 1.0' to 3.0'	
ST-50	For duct widths of 3.0' to 5.0	(requires support)
ST-100*	For duct widths of 5.0' to 10.0'	(requires support)

* This model is supplied as two 5 ft. sections with a coupling. Assembly is required for installation.

NOTE

Each model is manufactured with a different number and size of sampling holes. Only the specific tube must be used for the specified duct width.

Standard sampling tubes are steel tubes with air intake holes drilled the entire length of the tube. These tubes can be cut to length and must span at least 80% the width of the duct. Sampling tubes over 3.0' must be supported on the opposite side of the duct. To ensure the correct operation, the red stopper (stopper in installation kit) must be inserted in the end of the air input sampling tube. If the input tube protrudes through the opposite side of the duct, the opening around the tube must be

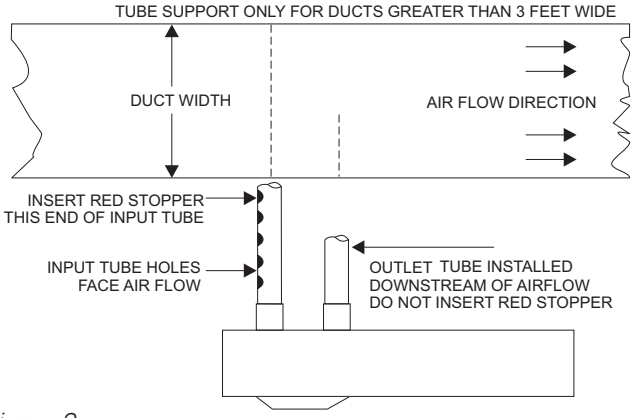
sealed. For custom duct widths, always use the next longest standard size and cut down to the exact requirement.

Once the airflow direction has been determined (refer to Figure 3), insert the input and outlet tubes into the duct housing.

1. Remove the cover from the housing.
2. Loosen the screw and rotate the tube retainer until the input tube is inserted and oriented properly. Ensure that the notched end of the tube is inside the housing and that the air input sampling tube is positioned so that the input holes are directly facing the airflow.
3. Once the tube is installed, rotate the retainer back into place and tighten screw.
4. Install the outlet tube in the remaining position. Once the tube is installed, rotate the retainer back into place and tighten screw.

Mounting

After securing the input and outlet tubes to the duct smoke unit, (or initially placing the tubes through the 1¼" holes drilled or punched in the HVAC duct to accept the input and outlet tubes and then attaching them to the duct unit), hold the duct unit assembly in position and use (2) # 12 X 3/4" sheet metal screws (packaged in the installation kit) to secure the duct smoke detector to the HVAC duct sheet metal.



NOTE: Mountings shown are typical. Detectors can be installed side, bottom or top of duct as long as proper tube operation and flow/pressure performance is maintained.

Figure 3
Sampling Tube Orientation

Air Duct Sampling Tube Pressure Measurement

The Model PDM-3 Pressure Differential Measuring device should be used to ensure that the sampling tube pressure differential is within the specified limits. The differential pressure between the two tubes should be greater than 0.01 inches of water and less than 1.2 inches of water. Qualified personnel should take measurements in accordance with the PDM-3 instructions, P/N 315-085535.

WIRING

Conduit Knockouts

To remove plastic knockout, first cut the plastic tab on the specific knockout. Refer to Figure 4 to locate the tab. Then, using a screw driver placed at the tab location, tap with a hammer until the knockout breaks out. Clean the hole before installing conduit.

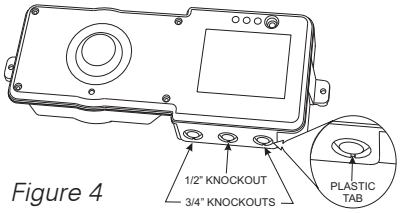


Figure 4
Removing the Knockouts

Wiring

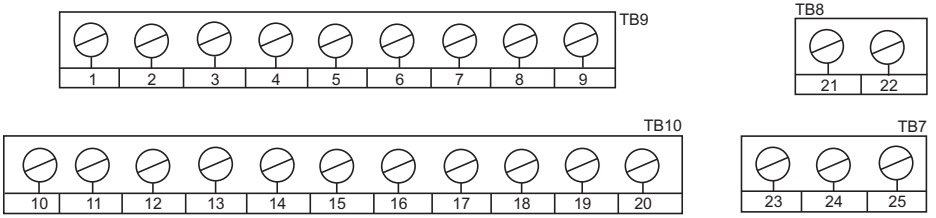
FDBZ492-PR terminal wiring connections are shown in Figure 5. *Note any limitations on the number of detectors and restrictions on the use of remote devices permitted for each circuit.*



Do not use looped wire under terminals. Break wire run to provide for proper supervision of connections.

With detector head removed, connect one of the appropriate dedicated power sources to the applicable terminals (Refer to Figure 5). Replace detector head and depress the cover removal switch (SW1) on the PCB assembly and the unit will be energized. The green power LED will be illuminated, and when pressing the test/reset button (SW2) on the PCB assembly, the red alarm LED will be illuminated. This test confirms the correct basic operation of the duct smoke unit, excluding the detector head (see functional testing).

Wire the FDBZ492-PR as a stand alone unit as shown in Figure 6.
Wire the FDBZ492-PR as part of a FACP as shown in Figure 7.
Wire the FDBZ492-PR accessories as shown in Figures 7 and 8.



FDBZ492-PR WIRING				
TERMINAL	SILK SCREEN MARKING	FUNCTION		CONNECTION
TB9	1	ALARM RELAY 1	NC	Alarm 1 relay - NC contact - Opened when smoke detected
	2		COM	Alarm 1 relay - COM contact
	3		NO	Alarm 1 relay - NO contact - Closed when smoke detected
	4	ALARM RELAY 2	NC	Alarm 2 relay - NC contact - Opened when smoke detected
	5		COM	Alarm 2 relay - COM contact
	6		NO	Alarm 2 relay - NO contact - Closed when smoke detected
	7	TROUBLE	NC	Trouble relay - NC contact - Closed in normal standby mode
	8		COM	Trouble relay - COM contact
	9		NO	Trouble relay - NO contact - Open in normal standby mode
TB10	10	INTERCONNECT / REMOTE ACCESSORIES	RTL-CNT	Test alarm - Apply 24VDC to test
	11		TB+	Low current trouble output for remotes - +24VDC when unit detects trouble
	12		C-/RTL-	Ground
	13		C+/RTL+	24VDC output
	14		AL1	High current alarm output - +24VDC when smoke detected
	15		AL+	Low current alarm output for remotes - +24VDC when smoke detected
	16		P+	Power out to remotes - Cut when head or cover removed
	17		T/R+	Test/Reset input - Apply 24VDC to test or reset
	18		C+	24VDC output
	19	FACP	COM	Fire Alarm Control Panel Relay - COM contact
	20		NO	Fire Alarm Control Panel Relay - NO contact - Open in normal standby mode
TB8	21	INPUT POWER	24AC/DC	Power input - 24VAC Hot or 24VDC positive
	22		N/O	Power input - 24VAC Neutral or 24VDC negative
TB7	23		N	Power input - AC Neutral
	24		115	Power input - 115VAC Hot
	25		230	Power input - 230VAC Hot

Figure 5
Wiring the FDBZ492-PR

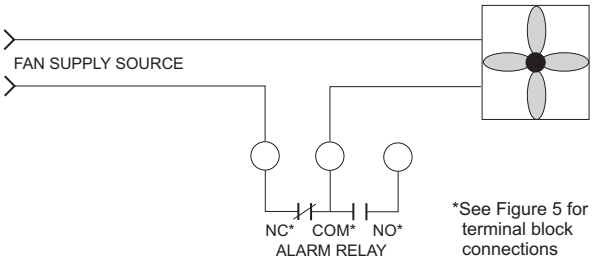
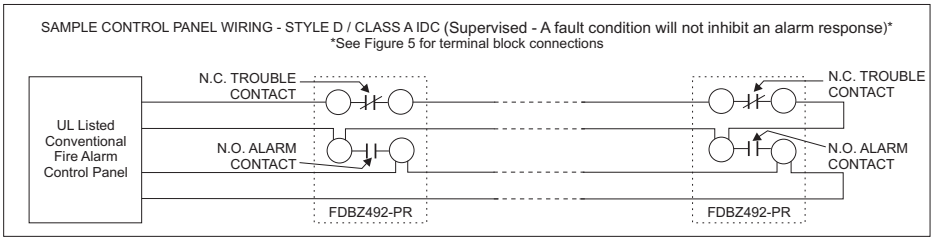


Figure 6
Stand Alone Wiring the FDBZ492-PR



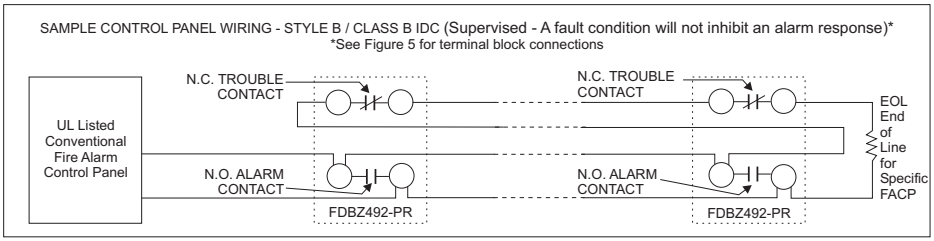


Figure 7
Wiring the FDBZ492-PR to a Fire Alarm Control Panel

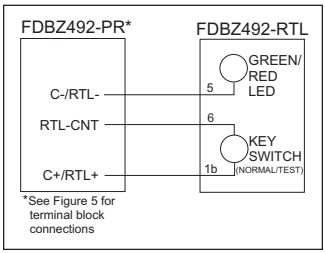
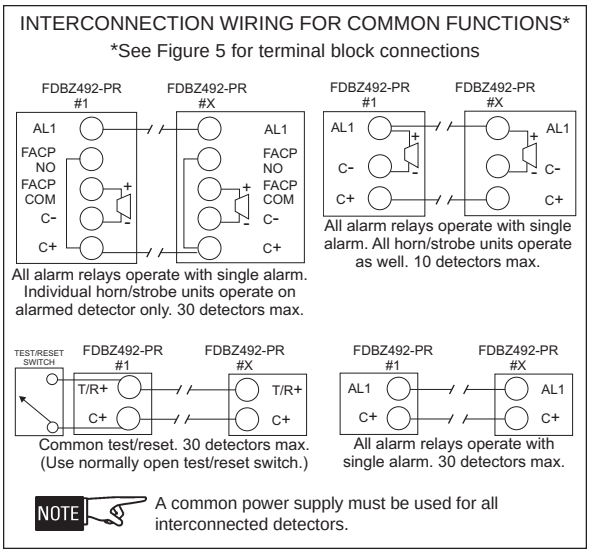


Figure 8
FDBZ492-PR Remote Accessory Wiring



In the event of a fire alarm, certain equipment may be required to be shut down. For example, shut down may be achieved by interrupting the fan supply source to that particular piece of equipment when wired as indicated in Figure 6.

Installation Of Smoke Detectors

To Install:

- Remove cover by loosening the four screws. Take off the cover and set it aside.
- Align detector with base and insert detector.
- Rotate detector clockwise while gently pressing on it until the detector drops fully into base.
- Then rotate the detector clockwise until it stops and snaps in place.
- Replace cover and tighten the four screws.

To Remove:

- Rotate the detector counterclockwise until stop is reached.
- Pull detector out of base.

TESTING

Only qualified service personnel should test these units. To assure proper operation of the detector and control panel, both the Sensitivity and the Functional tests should be conducted. The minimum test schedule may be found in the current edition of NFPA 72 for installations in the U.S

Sensitivity Test

The detectors monitor their smoke sensitivity automatically and require no test equipment. A green flash of the detector LED or in the FDBZ492-RTL Remote Accessory indicates that the smoke sensitivity is within its listed limits.

Functional Test

Smoke Testing

Using TG-11 smoke test canister with testing nozzle model AD-TGN (purchased separately) available from Siemens Industry, Inc., insert the test gas nozzle into the hole in the red plug in the unit cover. Press can against cover for about 1/2 second to release gas into the chamber.



Do not spray gas for more than ½ second. Overuse of test gas may result in detector contamination.

After 15 to 20 seconds the detector will go into alarm, illuminating the detector LED and causing the duct unit functions to operate; alarm relays will change state, and the alarm related remote accessories, if attached, will function.

If no test gas is available to conduct functional testing, remove cover and, while holding down the cover removal switch, blow smoke from a smoldering cotton wick or punk directly at the head to cause an alarm. The alarm indicator on the detector should illuminate within one minute.

Refer to the PE-11 Installation Instructions, P/N 315-094198, the PE-11C Installation Instructions, P/N 315-095626, the 8854 Installation Instructions, P/N 315-094198FA, or the OP121 Installation Instructions, P/N A6V10281367, as applicable, for additional information on testing these detectors.

MAINTENANCE

This unit is equipped with cover removal switch (SW1) that instantly provides a trouble condition upon removal of the clear cover. For all testing and inspection with the cover removed, the cover removal switch (designated as SW1 on PCB) must be manually depressed to simulate normal operation.

The performance of the air duct detector unit may be adversely affected by dirt or foreign matter on the sampling tubes or detector. If the air holes in the input sampling tube become restricted, the unit cannot receive a proper air sample, and performance is impaired. It is recommended that the sampling tubes be checked and cleaned periodically. The detector maintenance program should consist of periodic cleaning of dust from the detector head by using a vacuum cleaner. For cleaning, refer to the PE-11 Installation Instructions, P/N 315-094198, the PE-11C Installation Instructions, P/N 315-095626, or the 8854 Installation Instructions, P/N 315-094198FA.



Under no circumstances is the OP121 detector head to be disassembled. No repairs or cleaning should be attempted.

The cleaning and test program is recommended for 6 month intervals, or more frequently, if needed, depending on the individual detector environment. Consult your local code and AHJ requirements for required maintenance schedules.



Under no circumstances is the detector portion of the PE-11, PE-11C or 8853 to be disassembled by anyone other than an authorized Siemens Industry, Inc. Systems Technician. For service, contact your nearest authorized Siemens Industry, Inc. Service Representative.



If the fire alarm system is connected to a central station or fire department, or operates external devices such as fans, extinguishers, etc., connected, notify appropriate personnel and disconnect the external devices until all tests are completed. After testing, reset the system, reconnect the devices, and notify the personnel that the system is operating again.

ELECTRICAL RATINGS

PE-11, PE-11C AND 8854	
Voltage	16-27 VDC
Ripple	3V peak-to-peak
Supervisory Current	110uA max.
Start-up Time	50 seconds max.
Alarm Current	33 - 50mA

OP121	
Voltage	16-27 VDC
Ripple	3V peak-to-peak
Supervisory Current	110uA max.
Start-up Time	30 seconds max.
Alarm Current	30 - 50mA

FDBZ492-PR			
INPUT CURRENT	STANDBY CURRENT (Typical)	ALARM CURRENT (Typical)	TROUBLE CURRENT (Typical)
24VDC	29mA	123mA	20mA
24VAC @ 60Hz	80mA	273mA	65mA
115VAC @ 60Hz	26mA	57mA	35mA
230VAC @ 60Hz	30mA	38mA	20mA

COMPATIBLE CONTROL EQUIPMENT

FDBZ492-PR USING PE-11/-11C/8854/OP121 DETECTOR

Module Equipment Compatibility Identifier	Module Installation/ Wiring Instructions
CDC-4 (FireFinder-XLS)	P/N 315-034100-8
CP-35 (SYSTEM 3)	P/N 315-084902-22
*CZM-1B6 (MXL, MXL-IQ)	P/N 315-095355-9
CZM-4 (MXL, MXL-IQ)	P/N 315-090826-10
HZM (FireFinder-XLS, FS-250, FC2050, FC2025, FC2005, FC922, FC924, FC901)	P/N 315-034850-6
SXL-EX	P/N 315-095997-8
SZE-4X (SXL-EX)	P/N 315-096018-8
SZE-8AX (SXL-EX)	P/N 315-096022-8
*SZM (FS-250C)	P/N 315-034850C-4
ZU-35 (SYSTEM 3)	P/N 315-083222-18
8705 (MPC-6000/MPC-7000)	P/N 315-447309-8

REMOTE ACCESSORIES

The accessories available for use with FDBZ492-PR are:

REMOTE ACCESSORIES						ELECTRICAL RATINGS		
PART NUMBER/MODEL	DESCRIPTION	PILOT LED (Green)	ALARM LED (Red)	TROUBLE LED (Yellow)	HORN	STANDBY	ALARM	TROUBLE
S54319-S27-A1/FDBZ492-RTL	Remote Alarm, Pilot, key-operated Test/Reset Switch	•	•			0.2mA	0.2mA	0mA

SMOKE TEST GAS	
500-649750	TG-11 Detector Test Gas
500-649717	AD-TGN Test Gas Nozzle Kit

SMOKE DETECTOR HEAD	
500-094150	PE-11 Photoelectric Smoke Detector
500-095630	PE-11C Photoelectric Smoke Detector
500-094150FA	8854 Photoelectric Smoke Detector
S54372-F1-A1	OP121 Photoelectric Smoke Detector

WATERTIGHT HOUSING	
S543-B26-A1	FDBZ492-WT

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